```
!ls
```

```
'brain_tumor_dataset (1).zip'   dataset       test.jpg.jpeg
 brain_tumor_dataset.zip        sample_data
```

```
from google.colab import files
uploaded = files.upload()
! unzip -o brain_tumor_dataset.zip
!mkdir dataset
!mv yes dataset/
!mv no dataset/
import tensorflow as tf
import matplotlib.pyplot as plt
from tensorflow.keras.preprocessing.image import ImageDataGenerator

dataset_path = "/content/dataset"

datagen = ImageDataGenerator(
    rescale=1./255,
    validation_split=0.2
)

train_data = datagen.flow_from_directory(
    dataset_path,
    target_size=(128,128),
    batch_size=32,
    class_mode='binary',
    subset='training'
)

val_data = datagen.flow_from_directory(
    dataset_path,
    target_size=(128,128),
    batch_size=32,
    class_mode='binary',
    subset='validation'
)
print("Training samples:", train_data.samples)
print("Validation samples:", val_data.samples)
print("Classes:", train_data.class_indices)
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten,
Dense

model = Sequential([
    Conv2D(32,(3,3),activation='relu',input_shape=(128,128,3)),
    MaxPooling2D(2,2),

    Conv2D(64,(3,3),activation='relu'),
    MaxPooling2D(2,2),
```

```
    Conv2D(128,(3,3),activation='relu'),
    MaxPooling2D(2,2),

    Flatten(),
    Dense(128,activation='relu'),
    Dense(1,activation='sigmoid')
])

model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)

history = model.fit(
    train_data,
    epochs=10,
    validation_data=val_data
)

loss, accuracy = model.evaluate(val_data)
print("Validation Accuracy:", accuracy)

from tensorflow.keras.preprocessing import image
import numpy as np
import matplotlib.pyplot as plt
from tensorflow.keras.models import Model

# Load image
img_path = "/content/test.jpg"

img = image.load_img(img_path, target_size=(128,128))
img_array = image.img_to_array(img)/255.0
img_array = np.expand_dims(img_array, axis=0)

# Get conv layer outputs
layer_outputs = [layer.output for layer in model.layers if 'conv' in
layer.name]

# FIXED LINE
activation_model = Model(inputs=model.inputs, outputs=layer_outputs)

activations = activation_model.predict(img_array)

# Plot
for i, activation in enumerate(activations):
    plt.figure(figsize=(10,5))

    for j in range(min(6, activation.shape[-1])):
```

```
        plt.subplot(1,6,j+1)
        plt.imshow(activation[0,:,:,j], cmap='viridis')
        plt.axis('off')

    plt.suptitle(f"Layer {i+1} Feature Maps")
    plt.show()
img = image.load_img("/content/test.jpg", target_size=(128,128))
img_array = image.img_to_array(img)/255.0
img_array = np.expand_dims(img_array, axis=0)

prediction = model.predict(img_array)

if prediction[0] > 0.5:
    print("Tumor Detected")
else:
    print("No Tumor")
import tensorflow as tf
import cv2
import numpy as np
import matplotlib.pyplot as plt

from tensorflow.keras.models import Model
import tensorflow as tf
import numpy as np

def get_gradcam(model, img_array, last_conv_layer_name):

    # Get last conv layer
    last_conv_layer = model.get_layer(last_conv_layer_name)

    # Create a model that maps input → last conv layer output
    conv_model = tf.keras.Model(
        inputs=model.inputs,
        outputs=last_conv_layer.output
    )

    # Create classifier model (after conv layer)
    classifier_input =
tf.keras.Input(shape=last_conv_layer.output.shape[1:])

    x = classifier_input
    for layer in model.layers[model.layers.index(last_conv_layer)+1:]:
        x = layer(x)

    classifier_model = tf.keras.Model(classifier_input, x)

    # Gradient computation
    with tf.GradientTape() as tape:
        conv_outputs = conv_model(img_array)
        tape.watch(conv_outputs)
```

```
        predictions = classifier_model(conv_outputs)
        class_idx = tf.argmax(predictions[0])
        loss = predictions[:, class_idx]

    grads = tape.gradient(loss, conv_outputs)

    pooled_grads = tf.reduce_mean(grads, axis=(0,1,2))

    conv_outputs = conv_outputs[0]

    heatmap = conv_outputs @ pooled_grads[..., tf.newaxis]
    heatmap = tf.squeeze(heatmap)

    heatmap = tf.maximum(heatmap, 0) / tf.reduce_max(heatmap)

    return heatmap.numpy()
# Use SAME test image
img_path = "/content/test.jpg"

img = image.load_img(img_path, target_size=(128,128))
img_array = image.img_to_array(img)/255.0
img_array = np.expand_dims(img_array, axis=0)
_ = model.predict(img_array)
# IMPORTANT: change if different
last_conv_layer_name = None

for layer in reversed(model.layers):
    if isinstance(layer, tf.keras.layers.Conv2D):
        last_conv_layer_name = layer.name
        break

print("Using layer:", last_conv_layer_name)

heatmap = get_gradcam(model, img_array, last_conv_layer_name)
# Load original image using cv2
img = cv2.imread(img_path)
img = cv2.resize(img, (128,128))

heatmap = cv2.resize(heatmap, (128,128))
heatmap = np.uint8(255 * heatmap)

heatmap = cv2.applyColorMap(heatmap, cv2.COLORMAP_JET)

# Superimpose
superimposed_img = heatmap * 0.4 + img

# Show result
plt.imshow(cv2.cvtColor(superimposed_img.astype('uint8'),
cv2.COLOR_BGR2RGB))
plt.title("Grad-CAM (Tumor Region)")
```

```
plt.axis('off')
plt.show()
plt.plot(history.history['accuracy'], label='train')
plt.plot(history.history['val_accuracy'], label='validation')

plt.title("Model Accuracy")
plt.legend()
plt.show()
```

```
<IPython.core.display.HTML object>
```

```
Saving brain_tumor_dataset.zip to brain_tumor_dataset (2).zip
Archive:  brain_tumor_dataset.zip
  inflating: no/1 no.jpeg
  inflating: no/10 no.jpg
  inflating: no/11 no.jpg
  inflating: no/12 no.jpg
  inflating: no/13 no.jpg
  inflating: no/14 no.jpg
  inflating: no/15 no.jpg
  inflating: no/17 no.jpg
  inflating: no/18 no.jpg
  inflating: no/19 no.jpg
  inflating: no/2 no.jpeg
  inflating: no/20 no.jpg
  inflating: no/21 no.jpg
  inflating: no/22 no.jpg
  inflating: no/23 no.jpg
  inflating: no/24 no.jpg
  inflating: no/25 no.jpg
  inflating: no/26 no.jpg
  inflating: no/27 no.jpg
  inflating: no/28 no.jpg
  inflating: no/29 no.jpg
  inflating: no/3 no.jpg
  inflating: no/30 no.jpg
  inflating: no/31 no.jpg
  inflating: no/32 no.jpg
  inflating: no/33 no.jpg
  inflating: no/34 no.jpg
  inflating: no/35 no.jpg
  inflating: no/36 no.jpg
  inflating: no/37 no.jpg
  inflating: no/38 no.jpg
  inflating: no/39 no.jpg
  inflating: no/4 no.jpg
  inflating: no/40 no.jpg
  inflating: no/41 no.jpg
  inflating: no/42 no.jpg
  inflating: no/43 no.jpg
```

```
  inflating: no/44no.jpg
  inflating: no/45 no.jpg
  inflating: no/46 no.jpg
  inflating: no/47 no.jpg
  inflating: no/48 no.jpeg
  inflating: no/49 no.jpg
  inflating: no/5 no.jpg
  inflating: no/50 no.jpg
  inflating: no/6 no.jpg
  inflating: no/7 no.jpg
  inflating: no/8 no.jpg
  inflating: no/9 no.jpg
  inflating: no/N1.JPG
  inflating: no/N11.jpg
  inflating: no/N15.jpg
  inflating: no/N16.jpg
  inflating: no/N17.jpg
  inflating: no/N19.JPG
  inflating: no/N2.JPG
  inflating: no/N20.JPG
  inflating: no/N21.jpg
  inflating: no/N22.JPG
  inflating: no/N26.JPG
  inflating: no/N3.jpg
  inflating: no/N5.jpg
  inflating: no/N6.jpg
  inflating: no/No11.jpg
  inflating: no/No12.jpg
  inflating: no/No13.jpg
  inflating: no/No14.jpg
  inflating: no/No15.jpg
  inflating: no/No16.jpg
  inflating: no/No17.jpg
  inflating: no/No18.jpg
  inflating: no/No19.jpg
  inflating: no/No20.jpg
  inflating: no/No21.jpg
  inflating: no/No22.jpg
  inflating: no/no 1.jpg
  inflating: no/no 10.jpg
  inflating: no/no 100.jpg
  inflating: no/no 2.jpg
  inflating: no/no 3.jpg
  inflating: no/no 4.jpg
  inflating: no/no 5.jpeg
  inflating: no/no 6.jpg
  inflating: no/no 7.jpeg
  inflating: no/no 8.jpg
  inflating: no/no 89.jpg
```

```
  inflating: no/no 9.png
  inflating: no/no 90.jpg
  inflating: no/no 91.jpeg
  inflating: no/no 92.jpg
  inflating: no/no 923.jpg
  inflating: no/no 94.jpg
  inflating: no/no 95.jpg
  inflating: no/no 96.jpg
  inflating: no/no 97.jpg
  inflating: no/no 98.jpg
  inflating: no/no 99.jpg
  inflating: no/no.jpg
  inflating: yes/Y1.jpg
  inflating: yes/Y10.jpg
  inflating: yes/Y100.JPG
  inflating: yes/Y101.jpg
  inflating: yes/Y102.jpg
  inflating: yes/Y103.jpg
  inflating: yes/Y104.jpg
  inflating: yes/Y105.jpg
  inflating: yes/Y106.jpg
  inflating: yes/Y107.jpg
  inflating: yes/Y108.jpg
  inflating: yes/Y109.JPG
  inflating: yes/Y11.jpg
  inflating: yes/Y111.JPG
  inflating: yes/Y112.JPG
  inflating: yes/Y113.JPG
  inflating: yes/Y114.JPG
  inflating: yes/Y115.JPG
  inflating: yes/Y116.JPG
  inflating: yes/Y117.JPG
  inflating: yes/Y12.jpg
  inflating: yes/Y120.JPG
  inflating: yes/Y13.jpg
  inflating: yes/Y14.jpg
  inflating: yes/Y146.JPG
  inflating: yes/Y147.JPG
  inflating: yes/Y148.JPG
  inflating: yes/Y15.jpg
  inflating: yes/Y153.jpg
  inflating: yes/Y154.jpg
  inflating: yes/Y155.JPG
  inflating: yes/Y156.JPG
  inflating: yes/Y157.JPG
  inflating: yes/Y158.JPG
  inflating: yes/Y159.JPG
  inflating: yes/Y16.JPG
  inflating: yes/Y160.JPG
```

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  inflating: yes/Y161.JPG
  inflating: yes/Y162.jpg
  inflating: yes/Y163.JPG
  inflating: yes/Y164.JPG
  inflating: yes/Y165.JPG
  inflating: yes/Y166.JPG
  inflating: yes/Y167.JPG
  inflating: yes/Y168.jpg
  inflating: yes/Y169.jpg
  inflating: yes/Y17.jpg
  inflating: yes/Y170.JPG
  inflating: yes/Y18.JPG
  inflating: yes/Y180.jpg
  inflating: yes/Y181.jpg
  inflating: yes/Y182.JPG
  inflating: yes/Y183.jpg
  inflating: yes/Y184.JPG
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  inflating: yes/Y187.jpg
  inflating: yes/Y188.jpg
  inflating: yes/Y19.JPG
  inflating: yes/Y192.JPG
  inflating: yes/Y193.JPG
  inflating: yes/Y194.jpg
  inflating: yes/Y195.JPG
  inflating: yes/Y2.jpg
  inflating: yes/Y20.jpg
  inflating: yes/Y21.jpg
  inflating: yes/Y22.jpg
  inflating: yes/Y23.JPG
  inflating: yes/Y24.jpg
  inflating: yes/Y242.JPG
  inflating: yes/Y243.JPG
  inflating: yes/Y244.JPG
  inflating: yes/Y245.jpg
  inflating: yes/Y246.JPG
  inflating: yes/Y247.JPG
  inflating: yes/Y248.JPG
  inflating: yes/Y249.JPG
  inflating: yes/Y25.jpg
  inflating: yes/Y250.jpg
  inflating: yes/Y251.JPG
  inflating: yes/Y252.jpg
  inflating: yes/Y253.JPG
  inflating: yes/Y254.jpg
  inflating: yes/Y255.JPG
  inflating: yes/Y256.JPG
  inflating: yes/Y257.jpg
```

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  inflating: yes/Y258.JPG
  inflating: yes/Y259.JPG
  inflating: yes/Y26.jpg
  inflating: yes/Y27.jpg
  inflating: yes/Y28.jpg
  inflating: yes/Y29.jpg
  inflating: yes/Y3.jpg
  inflating: yes/Y30.jpg
  inflating: yes/Y31.jpg
  inflating: yes/Y32.jpg
  inflating: yes/Y33.jpg
  inflating: yes/Y34.jpg
  inflating: yes/Y35.jpg
  inflating: yes/Y36.JPG
  inflating: yes/Y37.jpg
  inflating: yes/Y38.jpg
  inflating: yes/Y39.jpg
  inflating: yes/Y4.jpg
  inflating: yes/Y40.JPG
  inflating: yes/Y41.jpg
  inflating: yes/Y42.jpg
  inflating: yes/Y44.JPG
  inflating: yes/Y45.JPG
  inflating: yes/Y46.jpg
  inflating: yes/Y47.JPG
  inflating: yes/Y49.JPG
  inflating: yes/Y50.JPG
  inflating: yes/Y51.jpg
  inflating: yes/Y52.jpg
  inflating: yes/Y53.jpg
  inflating: yes/Y54.jpg
  inflating: yes/Y55.jpg
  inflating: yes/Y56.jpg
  inflating: yes/Y58.JPG
  inflating: yes/Y59.JPG
  inflating: yes/Y6.jpg
  inflating: yes/Y60.jpg
  inflating: yes/Y61.jpg
  inflating: yes/Y62.jpg
  inflating: yes/Y65.JPG
  inflating: yes/Y66.JPG
  inflating: yes/Y67.JPG
  inflating: yes/Y69.jpg
  inflating: yes/Y7.jpg
  inflating: yes/Y70.jpg
  inflating: yes/Y71.JPG
  inflating: yes/Y73.jpg
  inflating: yes/Y74.jpg
  inflating: yes/Y75.JPG
```

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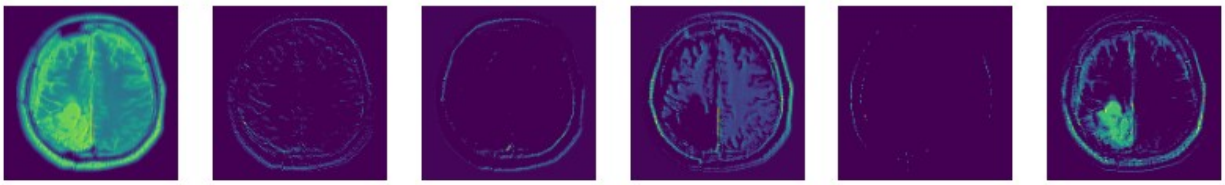
  inflating: yes/Y76.jpg
  inflating: yes/Y77.jpg
  inflating: yes/Y78.jpg
  inflating: yes/Y79.jpg
  inflating: yes/Y8.jpg
  inflating: yes/Y81.jpg
  inflating: yes/Y82.jpg
  inflating: yes/Y85.JPG
  inflating: yes/Y86.JPG
  inflating: yes/Y89.JPG
  inflating: yes/Y9.jpg
  inflating: yes/Y90.jpg
  inflating: yes/Y91.jpg
  inflating: yes/Y92.jpg
  inflating: yes/Y92.png
  inflating: yes/Y95.jpg
  inflating: yes/Y96.jpg
  inflating: yes/Y97.JPG
  inflating: yes/Y98.JPG
  inflating: yes/Y99.JPG
mkdir: cannot create directory ‘dataset’: File exists
mv: cannot move 'yes' to 'dataset/yes': Directory not empty
mv: cannot move 'no' to 'dataset/no': Directory not empty
Found 203 images belonging to 2 classes.
Found 50 images belonging to 2 classes.
Training samples: 203
Validation samples: 50
Classes: {'no': 0, 'yes': 1}
Epoch 1/10
7/7 ━━━━━━━━━━━━━━━━━━━━ 5s 651ms/step - accuracy: 0.6650 - loss:
0.7377 - val_accuracy: 0.7600 - val_loss: 0.5337
Epoch 2/10
7/7 ━━━━━━━━━━━━━━━━━━━━ 4s 628ms/step - accuracy: 0.7980 - loss:
0.4946 - val_accuracy: 0.7400 - val_loss: 0.5554
Epoch 3/10
7/7 ━━━━━━━━━━━━━━━━━━━━ 4s 548ms/step - accuracy: 0.8177 - loss:
0.4414 - val_accuracy: 0.7400 - val_loss: 0.5022
Epoch 4/10
7/7 ━━━━━━━━━━━━━━━━━━━━ 4s 551ms/step - accuracy: 0.8473 - loss:
0.3971 - val_accuracy: 0.7600 - val_loss: 0.5002
Epoch 5/10
7/7 ━━━━━━━━━━━━━━━━━━━━ 4s 631ms/step - accuracy: 0.8867 - loss:
0.3343 - val_accuracy: 0.7800 - val_loss: 0.5278
Epoch 6/10
7/7 ━━━━━━━━━━━━━━━━━━━━ 4s 549ms/step - accuracy: 0.8818 - loss:
0.3004 - val_accuracy: 0.8200 - val_loss: 0.4434
Epoch 7/10
7/7 ━━━━━━━━━━━━━━━━━━━━ 4s 603ms/step - accuracy: 0.8966 - loss:
0.2458 - val_accuracy: 0.7600 - val_loss: 0.4807
```

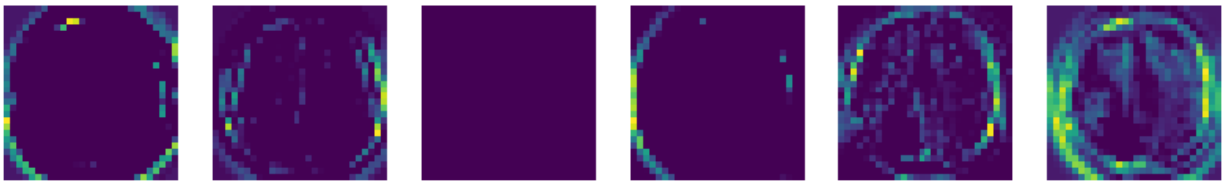
```
Epoch 8/10
7/7 ━━━━━━━━━━━━━━━━━━━━ 4s 627ms/step - accuracy: 0.9163 - loss:
0.2121 - val_accuracy: 0.7800 - val_loss: 0.7663
Epoch 9/10
7/7 ━━━━━━━━━━━━━━━━━━━━ 4s 548ms/step - accuracy: 0.8966 - loss:
0.2094 - val_accuracy: 0.8200 - val_loss: 0.4029
Epoch 10/10
7/7 ━━━━━━━━━━━━━━━━━━━━ 4s 551ms/step - accuracy: 0.9754 - loss:
0.1125 - val_accuracy: 0.8000 - val_loss: 0.6190
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 99ms/step - accuracy: 0.8000 - loss:
0.6190
Validation Accuracy: 0.800000011920929
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 98ms/step
```

Layer 1 Feature Maps



Layer 2 Feature Maps

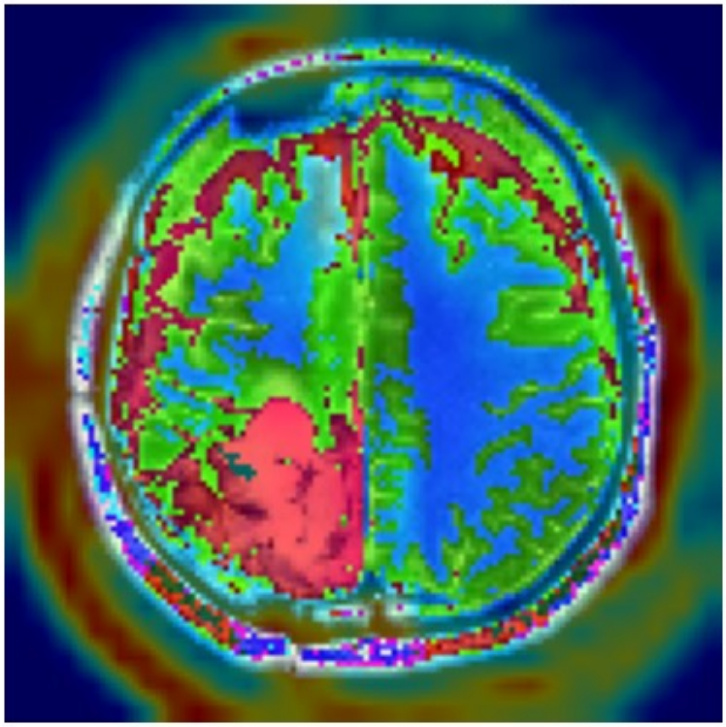
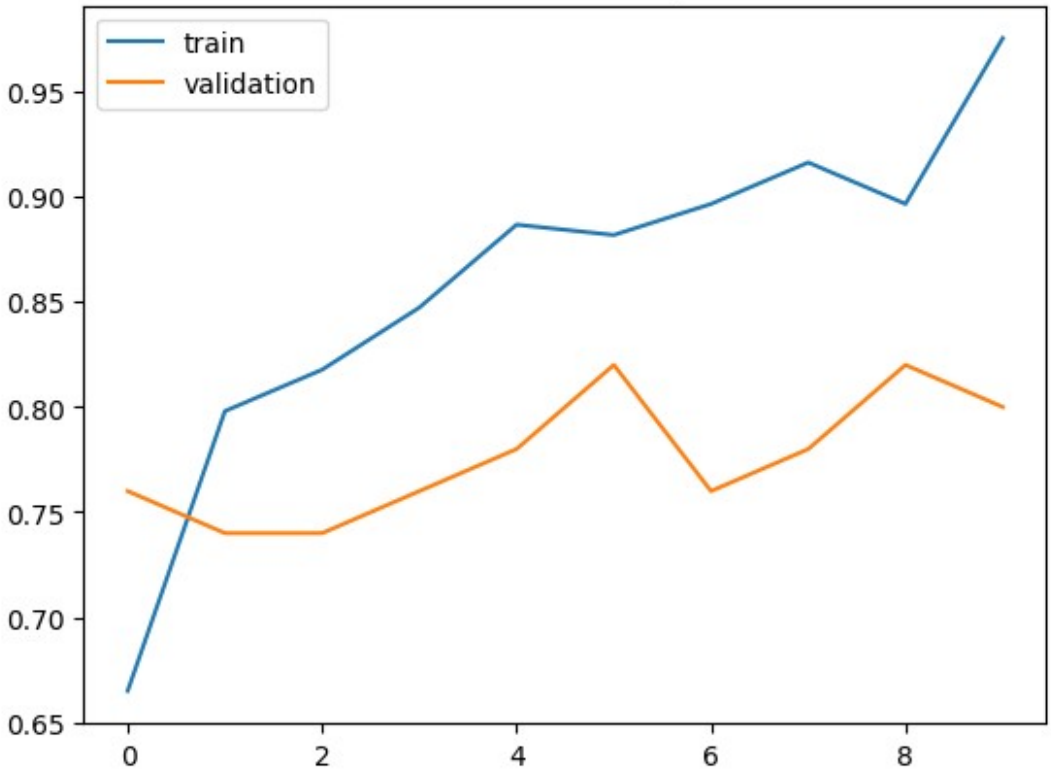
Layer 3 Feature Maps



```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 98ms/step
Tumor Detected
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 43ms/step
Using layer: conv2d_5

/usr/local/lib/python3.12/dist-packages/keras/src/models/
functional.py:241: UserWarning: The structure of `inputs` doesn't
match the expected structure.
Expected: ['keras_tensor_16']
Received: inputs=Tensor(shape=(1, 128, 128, 3))
  warnings.warn(msg)
```

Grad-CAM (Tumor Region)



Model Accuracy